

Project Title: Bank Vault Attack Challenge (Phase 1)

Task:

Develop a secure, multi-user Command Line Interface (CLI) banking executable **using C**. This system will serve as the core interface on which other teams try to find exploits, representing the first layer of the Bank Vault architecture.

Functional Requirements:

1. User Management & Authentication

- **Multi-Account Support:** The system must support multiple distinct accounts.
- **Account Separation:** Strict logical isolation must be enforced between accounts; users should not be able to access or modify another user's data.
- **Authentication:** Critical operations (specifically withdrawals) must be secured via a Personal Identification Number (PIN).
- **PIN Management:** Users must have the ability to change their PIN securely.

2. Banking Operations

- **Balance Query:** Users must be able to view their current account balance.
- **Withdrawals:** Users can withdraw funds, contingent upon successful PIN verification.
- **Loans:** Implement a facility for users to request and receive loans directly from the Admin account.

3. Fixed Deposit (FD) Module

- **Create FD:** Users can transfer a selected amount from their main balance into a Fixed Deposit.
 - *Validation:* The system must verify that the user has sufficient funds before creation.

- **Interest Calculation:** The FD balance must multiply based on a defined interest rate.
- **Break FD:** Users must be able to liquidate their FD, transferring the principal plus accumulated interest back into their main account balance.

4. Administrative Privileges

- **The ADMIN Account:** A superuser account ("ADMIN") must be implemented.
 - **Capabilities:** This account is unrestricted and must possess the ability to perform any operation on the system, including overriding user constraints.
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Technical & Security Constraints:

1. Sensitive Data Handling

- **Credit Card Integration:** The system must handle and store Credit Card Numbers (PAN) and CVV codes associated with accounts.

2. Delivery Platform

- The final deliverable must be a compiled **C executable**.
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Deliverables

- Source code in C.
- Compiled CLI Executable meeting all functional and security constraints listed above.